ViCo-LWIR: Vision-Based detection of Indoor Liquid Spills Using Long-Wave Infrared Imaging and a Weakly Supervised Contrastive Framework

Félix Olémou, Éric Hervet, and Ghazal Rouhafzay

***Abstract*—Liquid spill detection is important in safety-sensitive indoor environments, where undetected spills can damage equipment, disrupt operations, and increase accident risks. Visible-spectrum cameras often fail in this task because many liquids are transparent, irregularly shaped, surface-dependent, and sensitive to illumination conditions. This paper proposes ViCo-LWIR, a weakly supervised contrastive framework for image-level liquid spill detection using long-wave infrared (LWIR) imaging.**

The proposed method addresses the limited availability of densely annotated thermal spill datasets by learning discriminative representations from image-level labels. To handle the strong variability of thermal spill appearances, the framework incorporates uncertainty-aware positive weighting, hard negative mining, and curriculum learning. These components allow the model to preserve meaningful intra-class diversity while improving separation between spill and non-spill samples.

Experiments on LWIR spill images show that ViCo-LWIR outperforms several weakly supervised contrastive baselines in challenging indoor scenarios. Qualitative attention visualizations further indicate that the model focuses on relevant spill regions, although the evaluation is conducted primarily at the image level.
Link to code and dataset

***Index Terms*—Computer Vision; Long-Wave Infrared Sensing; Thermal Spill Detection; Contrastive learning; Learning under Uncertainty.**

I. Introduction

An effective and reliable spill detector is critical to the safety of physical infrastructures and indoor public environments. Undetected liquid leaks and spills can cause structural degradation, equipment failure, and cascading operational risks as well as safety hazards. Continuous monitoring is therefore essential for pipeline infrastructures [1]. Indoor spill detection is also important in smart city environments where undetected spills can increase slip risk and disrupt normal operations. Conventional visible spectrum sensors such as consumer and industrial Closed-Circuit Television (CCTV) cameras may fail in poorly lit environments [2]. Thermal imaging offers fundamental advantages for liquid spill detection by measuring emitted thermal radiation rather than reflected visible light. It provides a direct, non-contact measurement of surface temperature distributions, making them inherently robust to lighting variations, shadows, and low light conditions [3]. Furthermore, the distinct thermophysical properties of liquids give rise to surface temperature anomalies that make their presence observable in infrared thermal imagery. More specifically, liquids often induce measurable thermal anomalies due to differences in thermal conductivity, heat capacity, evaporation dynamics, and thermal inertia between liquids and solid surfaces, resulting in spatially coherent temperature patterns that can persist over time even when visible spectrum contrast is weak or absent [4]. Several studies have shown that such thermal signatures enable a clearer identification of leakage or spill regions compared to visible imagery, particularly in indoor or nighttime scenarios where RGB sensors suffer from noise amplification and contrast loss [5]. Building on these instrumentation advantages, deep learning can automate spill detection by learning discriminative patterns from thermal measurements and enabling continuous monitoring. Unlike visible-spectrum object detection, which benefits from large benchmark datasets and mature architectures, thermal object detection remains limited by smaller, less diverse datasets. This limitation is particularly challenging for liquid spill detection because spills have irregular, time-varying boundaries that depend on surface properties and environmental conditions. As a result, producing pixel-level masks or bounding boxes is not only labor-intensive but also subjective, especially when defining the exact extent of a spill in thermal imagery [6]. These annotation challenges make large-scale thermal datasets costly, time-consuming, and difficult to standardize. They also limit the use of fully supervised methods, motivating a weakly supervised framework that first detects whether a spill is present and then highlights the corresponding image regions. Weak supervision significantly simplifies the annotation charge, but introduces new learning challenges. Without dense spatial labels, the model must infer meaningful spatial and semantic structure from coarse supervisory signals. Under such conditions, learning objectives that rely on explicit class boundaries or precise localization may become unreliable. Contrastive learning is suitable in this context because it structures the embedding space by encouraging similarity between related samples and dissimilarity between unrelated ones, without requiring pixel-level annotation [7]. Recent works such as Learning reliable representations for liquid spill in thermal imagery is intrinsically challenging due to both geometric ambiguity and thermophysical variability. Unlike rigid objects with stable contours, liquid spills presents irregular and evolving spatial extents. In infrared images, their apparent shape and size are governed by temperature gradients, surface materials, and environmental conditions. As

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such, spill boundaries are inherently uncertain, and a single scene may contain mixed thermal responses that cannot be cleanly separated without pixel-level annotations.

Thermal spill detection is further complicated by high intra-class variability and ambiguous thermal responses. The same liquid may appear thermally bright or dark depending on its contrast with the background, while unrelated structures such as heated surfaces or residual thermal patterns may produce similar signatures. This ambiguity can cause learning algorithms to associate background artifacts with spill regions or suppress subtle but physically meaningful spill patterns.

Such ambiguity is particularly problematic for contrastive learning, which depends on reliable similarity and dissimilarity relationships in the embedding space. Methods such as SimCLR [8] and MoCo [9] assume that different images are semantically unrelated and should be separated. However, in thermal spill detection, different scenes or regions may reflect the same physical phenomenon. Strongly repelling them can therefore fragment the representation space and reduce its consistency with the underlying thermal process.

Several extensions have attempted to address this limitation. Debiased contrastive learning [10] reduces the impact of false negatives by reweighting the contrastive loss, but it does not resolve semantic uncertainty. Supervised contrastive learning [11] uses labels to enforce intra-class compactness, but it assumes that samples within a class share consistent visual patterns. This assumption is weak in thermal spill imagery, where the same liquid may appear thermally bright or dark, and unrelated hot surfaces may produce similar responses. Prototype-based contrastive learning [12] addresses intra-class variability by attracting samples toward class centroids, but a single prototype cannot represent the diverse thermal appearances of spills. Multi-prototype variants offer more flexibility, but introduce additional complexity and may be sensitive to prototype selection or collapse.

These limitations show that weakly supervised thermal spill detection requires a contrastive strategy that accounts for uncertain boundaries, strong intra-class variability, and physically induced ambiguity. This motivates our curriculum-based framework, which progressively structures the embedding space while preserving meaningful diversity among spill samples.

In view of all the points invoked, we aim to address these limitations and challenges by introducing a new contrastive learning framework. Our key contributions are as follows.

- we tackle intra-class variability by weighting samples according to their uncertainty. This allows samples that share the same semantic meaning in a class, despite being ambiguous or visually diverse, to still co-exist in the same cluster without pushing each other aside. As a result, various types of spills remain compact within a single large and cohesive cluster;
- The high inter-class similarity is handled through a weighting of negative samples based on their dissimilarity to positive counterparts, ensuring stronger embedding separation between opposite classes;
- To ensure stable learning, we introduce a progressive learning mechanism that facilitates more effective model training;

II. Related Work

A. Image Level spill detection

Rowlands et al. [13] introduced a multi-spectral vision system combining long-wave infrared (thermal) and near-infrared (depth) imaging is developed for real-time bleeding detection using deep learning. The framework enhances liquid detection through optimized thermal-aware annotations, cross-modal region matching, and data fusion to distinguish blood from other warm liquids. Experimental results demonstrate that enclosing entire liquid patches and integrating information from both imaging modalities improves detection accuracy and reliability across different surfaces and practical scenarios. Similarly a thermal imaging-based liquid spill detection framework is presented for health and safety applications, addressing the lack of large annotated thermal datasets. Three annotation-free approaches are evaluated: watershed-based image processing, weakly supervised detection using Gradient Class Activation Mapping (Grad-CAM), and unsupervised motion-guided salient object detection. Experimental results show that the conventional watershed-based method achieves reliable warm liquid spill detection and localization [14].

B. Contrastive Learning

In recent years, there has been a trend towards adopting contrastive learning style framework in representation learning. The core principle of the contrastive learning framework is to pull samples from positive pairs in an identical cluster and repulse the opposite apart. Additionally, because of its ability to be designed as class-agnostic or class-aware, and to be invariant to shape and position, it led to weakly supervised object detection. To learn more consistent feature representations of the ground truth proposal, Seo et al. [15] leverage supervised contrastive learning (SupCon) by weighting it with the confidence score of the instance for the MIL head, allowing a credible similarity threshold for object discovery. Zhang et al. [16] propose Negative Prototypes Guided Contrastive Learning (NPGC) for WSOD, to better distinguish between object types. A global feature bank was introduced to store both positive and negative prototypes with an improvement proposal representation mechanism of pseudo-label sampling combined with contrastive learning.

However, to the best of our knowledge, the current state-of-the-art does not provide a unified methodology for detecting objects with shape invariance specifically spilled liquids using weakly supervised contrastive learning. This gap clarifies our motivation to consider this vision task as a research area to explore.

III. Methodology

In this section we propose a contrastive learning framework capable of achieving semantically meaningful clusters and improving the reliability of feature representations in measurement systems. This approach mitigates visual ambiguity

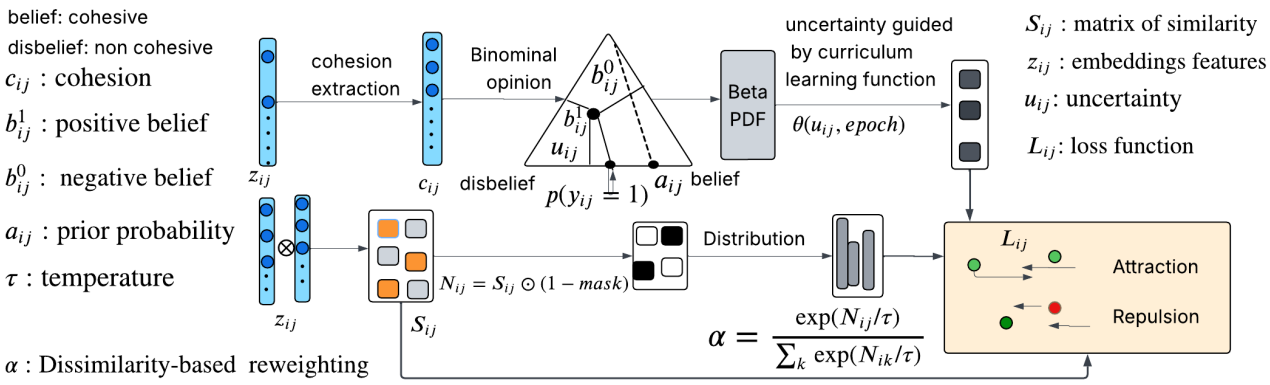


Fig. 1. Through the encoder, feature embeddings are extracted to obtain z_{ij}, which are subsequently normalized. From these embeddings, the cohesion score c_{ij} is computed, representing how likely samples x_i and x_j remain compact within the same co-cluster. A binomial opinion is modeled as a Beta probability density function (PDF), under the assumption of a bijective mapping between both representations. The uncertainty u_{ij} is then computed to measure the confidence that x_i and x_j can be close within a cluster, enabling the model to avoid forcing samples of the same class together despite strong visual differences. To ensure stable and gradual learning, a curriculum function (described in Section III) is introduced to guide progressive training and to compute the adaptive weight w_{ij}, addressing intra-class variability. For inter-class modeling, the parameter α is computed as described in the schema above, allowing the model to focus on hard negatives and enhance class separation. All these components are integrated into the final loss function $\mathcal{L}_{ij}$.

caused by low intra-class similarity and high inter-class similarity, enhancing the accuracy and robustness of automated instrumentation and sensor-based monitoring applications.

The following section is organized into four subsections. Section III-A describes the data acquisition process; Section III-B addresses learning under uncertainty; Section III-C presents the class-level separation strategy; and Section III-D introduces the progressive learning scheme. An overview of the proposed framework is illustrated in Fig. 1, and detailed descriptions are provided in the following subsections.

A. Long-Wave Infrared Imaging

Long-wave infrared (LWIR) imaging captures thermal radiation emitted by objects, providing a representation of scene temperature independent of visible illumination. LWIR imaging is particularly advantageous for tasks involving low visibility, material differentiation, or subtle surface phenomena that are difficult to detect in the visible spectrum. To capture these thermal characteristics in a compact form factor, the Lepton series from Teledyne FLIR is commonly employed in research. By shrinking advanced LWIR technology into a package smaller than a dime and being affordable at low cost, the FLIR Lepton 3.1R thermal sensor is the module used in the experiments conducted in this research. This compact sensor is suitable for applications such as hotspot detection in electrical systems, monitoring industrial facilities, identifying heat leaks in houses [17], and spill detection under low illumination.

The FLIR Lepton 3.1R thermal camera [18], adapted for night vision, offers a resolution of 160×120 pixels and a wide field of view of $95°$ horizontally and $119°$ diagonally. As a radiometric sensor, it provides an absolute temperature output with a thermal sensitivity of < 50 mK (0.050 °C) and can detect temperatures up to -10 °C to $+400$ °C (14 °F to 752 °F) at room temperature in low gain mode. A particular feature, a Non-Uniformity Correction (NUC) with an integral shutter, automatically corrects the sensor when the captured scene image quality is blurred.

The data acquisition was done as follows: The FLIR Lepton 3.1R long-wave infrared sensor captures the scene at a resolution of 160×120 pixels and approximately 9 fps. The Raw radiometric frames are converted to temperature values using factory-calibrated scaling. A fixed manual level span normalization is applied to preserve subtle thermal variations while limiting the influence of outliers, followed by linear normalization to an 8-bit grayscale representation. Contrast Limited Adaptive Histogram Equalization and controlled brightness scaling are employed to enhance low-contrast thermal structures while maintaining measurement consistency. Processed frames are recorded in real time for offline analysis, providing a stable and interpretable thermal representation suitable for learning under uncertainty and contrastive representation learning.

B. Learning under Uncertainty

Despite deep neural networks (DNNs) achieving remarkable success in making accurate predictions in computer vision tasks, they may sometimes make unexpected, inaccurate, and overconfident predictions due to uncertainty factors present in complex environments [19]. Uncertainty modeling can be grouped into aleatoric uncertainty and epistemic uncertainty. Data uncertainty(aleatoric uncertainty) refers to the inherent randomness and stochasticity of the data that originate from sensor noise. It is considered irreducible because, in fact, adding a surplus of training data does not bring any gain in performance. On the other hand, epistemic uncertainty or model uncertainty is caused by a lack of knowledge on the data [20]. To overcome these challenges, Bayesian Neural Networks (BNNs), considered as a standard formulation

for uncertainty quantification in neural networks, have been widely adopted in various methodologies that approximate Bayesian inference through techniques such as assembling or variational inference [21].

However, uncertainty from real-world data poses a major challenge for deep learning vision models which aim to partition the visual features embedding into distinctive and semantic clusters. Ambiguity in visual features can lead to a divergence in same-class clustering where samples from the same co-cluster appear dissimilar. This high intra-class variability accentuates the aleatoric uncertainty at the instance level, highlighting the challenge of correctly assigning samples to their cluster despite all belonging to the same class. Thereby we define the uncertainty inherent from pairwise similarity between samples from the same co-cluster as cohesive uncertainty. That is, the uncertainty about how well two instances from the same class can remain close or cohabit within the same cluster in the embedding space. We model the cohesive uncertainty to reduce the influence of uncertainty caused by visual data ambiguity through Subjective Logic (SL) theory [22].

1) Intra-class Uncertainty Modeling: In order to adapt Subjective Logic (SL) in a binary setting, we need to figure out beforehand how to obtain the cohesion c_{ij} of the instances x_i and x_j, such that they remain compact in a co-cluster. Furthermore, we define a multilayer perceptron (MLP) denoted as $\lambda_1(.)$ on top of a vision transformer base model as the projector head to map the representation of features X^f in the embedding subspace $z_i = \lambda_1(X^f)$. We also define s_{norm} pairwise normalized cosine similarities between embeddings in a batch of size $N = 64$ and embedding dimension where each embedding has dimension $d = 768$, matching the output of the Vision Transformer base model (ViT-B), and s_{ij} scalar normalized cosine similarity between embeddings z_i and z_j as follows:

$$\tilde{z} = \begin{bmatrix} \frac{z_1}{\|z_1\|_2} \\ \frac{z_2}{\|z_2\|_2} \\ \vdots \\ \frac{z_N}{\|z_N\|_2} \end{bmatrix} \in \mathbb{R}^{N\times d} \quad \text{and} \quad s_{\text{norm}} = \tilde{z}\,\tilde{z}^\top \in \mathbb{R}^{N\times N} \tag{1a}$$

$$s_{ij} = \frac{z_i z_j^\top}{\|z_i\|_2 \|z_j\|_2}, \quad i,j = 1,\ldots,N \tag{1b}$$

We formally define the cohesion c_{ij} between two instances x_i and x_j in a co-cluster as the exponentiated row-wise softmax of their cosine similarity. By applying the softmax function row-wise, the similarity of each sample to all other samples is normalized, highlighting how strongly it is associated with its neighboring samples in the cluster. Exponentiating the softmax amplifies the difference, however small, between strong and weak associations. This allows highly cohesive pairs to receive higher weights (approaching to e^1), while weaker cohesive ones are less emphasized; They receive values relatively closer to 1, which reduces their influence considerably. This ensures co-cluster compactness while preserving meaningful intra-class variability among sample. As for $\bar{c}_{ij}$, it denotes the non-cohesiveness between two instances x_i and x_j belonging to the same class. It quantifies the degree to which the pair is atypical or dispersed, reflecting how likely the two samples are to be non-compact despite sharing the same class membership.

$$\text{softmax}_j(s_{ij}) = \frac{\exp(s_{ij})}{\sum_{k=1}^{N} \exp(s_{ik})}, \quad i,j = 1,\ldots,N \tag{2a}$$

$$c_{ij} = \exp\Big(\text{softmax}_j(s_{ij})\Big), \quad i,j = 1,\ldots,N \tag{2b}$$

$$\bar{c}_{ij} = \exp\Big(\text{softmax}_j(1 - s_{ij})\Big), \quad i,j = 1,\ldots,N \tag{2c}$$

Our formulation assumes that pairwise cohesion in the learned embedding space provides evidence about the consistency of samples within a co-cluster. Under this assumption, uncertainty is computed from the amount of accumulated cohesion evidence rather than from the model's predicted class probabilities. Specifically, the ViT backbone and projector head generate embeddings (z_i), from which pairwise cosine similarities are computed and transformed into cohesion scores (c_{ij}). These cohesion scores constitute the evidence term in the Subjective Logic framework, and the corresponding belief and uncertainty quantities are obtained according to Eq.(4) and (5). Consequently, uncertainty decreases as supporting cohesion evidence increases and remains high when the evidence is weak or ambiguous. Since the uncertainty estimate is derived from pairwise relationships in the embedding space rather than from the final classifier output, the computation does not rely on a circular transformation of the model's own prediction confidence. Furthermore this formulation grounds uncertainty in the local structure of the learned embedding space and avoids deriving uncertainty as a post-hoc transformation of the model's prediction confidence.

Therefore a co-cluster there are two types of relationships between any two samples x_i and x_j 'cohesive' highly similar and 'non-cohesive intra-class sample' atypical, samples that are more distinctive despite sharing class membership. So, let m_{ij} be the proposition that expresses the relation of cohesiveness or non-cohesiveness. A binomial opinion of m_{ij} is represented by ϕ_{ij} = (b_{ij}, u_{ij}, a_{ij}) in a binary domain $\mathbb{D} = \{0, 1\}$, where:

- 1: designates a cohesive relation between x_i and x_j;
- 0: designates a non-cohesive relation between x_i and x_j;
- $\mathbf{u}_{ij}$: represents the cohesiveness uncertainty; i.e. the degree of vacuity of evidence;
- $\mathbf{b}_{ij}$: belief mass distribution over $\mathbb{D}$ for cohesiveness; $\mathbf{b}_{ij} = \begin{bmatrix} b_{ij}^0 \\ b_{ij}^1 \end{bmatrix}$
- $\mathbf{a}_{ij}$: base rate distribution over $\mathbb{D}$; $\mathbf{a}_{ij} = \begin{bmatrix} a_{ij}^0 \\ a_{ij}^1 \end{bmatrix}$.

Going forward, we assume that no instance is isolated by ensuring that we have at least two augmented views for a given anchor $anchor_i$ in a batch, which guarantees that each instance is clustered with at least one other instance. Moreover, when the negative opinion(non-cohesive intra-class sample) arises between any specific pair of instances indicating that they cannot be merged, this disbelief is not suppressed; Instead, it

is redistributed as a positive belief(cohesive) b_{ij}^1 across other intra-cluster pairs involving the same instance. This process highlights the cohesiveness of the cluster, as sub-clusters naturally form, thus reinforcing instance's membership. So, based on the assumption made and through a specific bijective mapping, a binomial opinion ϕ_{ij} is equivalent to a Beta PDF (probability density function) [23].

The projected probability distribution of the binomial opinion regardless of the cohesiveness or non-cohesiveness:

$$\Pr(m_{ij} = k) = b_{ij}^k + a_{ij}^k u_{ij}, \quad \forall k \in \mathbb{D}, \tag{3a}$$

$$\text{Beta}(p_{ij}; \beta_0, \beta_1) = \frac{\Gamma(\beta_0 + \beta_1)}{\Gamma(\beta_0)\Gamma(\beta_1)} p_{ij}^{\beta_0 - 1} \left(1 - p_{ij}^{\beta_1 - 1}\right), \tag{3b}$$

where $\beta_0 > 0$, $\beta_1 > 0$, $p_{ij} \neq 0$ if $\beta_0 < 1$, and $p_{ij} \neq 1$ if $\beta_1 < 1$, and the additivity requirement should hold with

$$\int_0^1 \text{Beta}(p_{ij}; \beta_0, \beta 1)\, dp_{ij} = 1.$$

$$\beta_0 = c_{ij} + a_{ij}^0 W, \quad \beta_1 = \bar{c}_{ij} + a_{ij}^1 W, \tag{3c}$$

The strength parameters β_0 and β_1 can simply be represented by the base rate a_{ij}^0 and the observation evidence $(c_{ij}, \bar{c}_{ij})$ where W is the prior weight or the number of times that we assume that an instance can be seen under a prior assumption.

Therefore, the Beta PDF has the probability Pr as a variable, where the expected probability over m_{ij} can be calculated as follows, considering Eq. (2b), (2c), (3a), (3b), and (3c):

$$E(m_{ij}) = \frac{\beta_0}{\beta_0 + \beta_1} = \frac{c_{ij} + a_{ij}^0 W}{c_{ij} + \bar{c}_{ij} + W}. \tag{4}$$

The equivalence between a binomial opinion and a Beta PDF can be achieved through the mapping defined below:

$$b_{ij}^0 = \frac{c_{ij}}{c_{ij} + \bar{c}_{ij} + W}, \quad u_{ij} = \frac{W}{c_{ij} + \bar{c}_{ij} + W}. \tag{5}$$

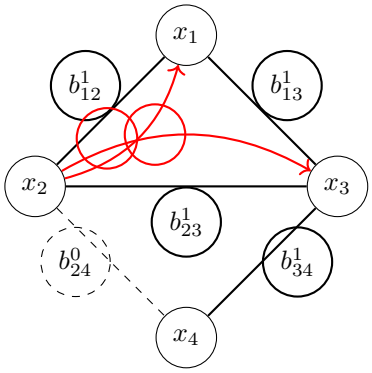


Fig. 2. Intra-cluster setup showing cohesiveness b_{ij}^1 (solid lines) strong intra-cluster similarity, non cohesive intra-class samples (more discriminative pair) b_{ij}^0 (dashed line), and redistribution of negative non cohesive intra-class samples (red arrows) forming cohesive sub-clusters.

The prior weight was set to $W = 2$ and the base rate to $a_{ij}^0 = 0.5$, following the standard formulation of subjective logic. This choice yields a uniform Beta(1,1) distribution when no evidence is available ($c_{ij} = \bar{c}_{ij} = 0$), thereby representing a non-informative and unbiased prior. Consequently, neither outcome is favored before observations are collected, while the uncertainty is progressively reduced as supporting and opposing evidence accumulates.

2) Progressive Uncertainty Weighting: In the early learning phase, most deep learning models in vision tasks are still learning relevant patterns or specific characteristics that are most meaningful for the task they were designed for. In this phase the learned features are uncertain, the model is not fully aware of what to learn or where to look for feature extraction. Adding to this constraint the effect of intra-class ambiguity which leads to a hard learning pattern due to the high uncertainty that occurs each time the model sees samples that are visually various despite sharing same class membership.

In terms of these constraints, it is relevant for us to mimic the progressive learning mechanism encountered through observation of the human way of learning. Humans tend during their learning process to start engaging with tasks that are relatively simple and then introduce hard ones gradually. This gradual learning process facilitates the understanding of a complex task that is subdivided into a series of easy tasks. In this context, we introduce a progressive learning function $\theta_{ij}(u_{ij}, epoch(i))$ where $epoch(i)$ denotes the current training epoch corresponding to iteration i. The function's role is to mimic curriculum learning. More details on this function are given in Section III-D. Let $\mathcal{A}$ denote the set of all augmented views derived from the instances $\{\mathbf{x}_k\}_{k=1}^N$ in the dataset $\mathcal{X}$. Let $orig(i)$ denote the index of the original (non-augmented) instance from which the augmented view $\mathbf{x}_i \in \mathcal{A}$ is derived, and let y_i denote the class label of that original instance. For any two augmented views $\mathbf{x}_i, \mathbf{x}_j \in \mathcal{A}$, the pairwise weighting coefficient w_{ij} is defined as follows:

$$w_{ij} = \begin{cases} 1, & \text{if } \big[\text{orig}(i) = \text{orig}(j)\big] \\ & \wedge \big[y_i = y_j\big] \wedge (i \neq j), \\ \theta_{ij}\big(u_{ij}, \text{epoch}(i)\big), & \text{if } \big[\text{orig}(i) \neq \text{orig}(j)\big] \\ & \wedge \big[y_i = y_j\big] \wedge (i \neq j), \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

This new weight w_{ij} introduced above is used in Section III-E as a weighting parameter to handle intra-class ambiguity which allows the model to be aware that two instances x_i and x_j could be visually different instead of pushing it apart, the weight adjusts their contribution appropriately.

C. Handling Inter-Class Variability

One known limitation of supervised contrastive learning is the high similarity between a given anchor and the negative sample present in a batch. This limitation can lead to a gradient confusion because the model is forced to repel two samples that are actually similar. Moreover, the loss may not decrease smoothly due to the fact that the model receives contradictory signals. On the one hand, it pushes similar samples; on the other hand, it pulls similar samples together to meet intra-class consistency. To mitigate this, we introduce a weighting parameter α, independent of the number of negative samples

present in a batch, which accounts for the relative position of negatives compared to others. This leads to a higher weight assignment to negatives that look highly similar, known as hard negatives, to an anchor, whereas dissimilar negatives receive lower weights. In this context, the model learns to focus on the hard negatives which emphasize the learning signal, encouraging smooth convergence and class separation. Using Eq.(1b) and the indicator function $\mathbf{1}_{[y_i \neq y_k]}$ that equals 1 if the condition is true and 0 otherwise. we have:

$$\alpha_{ij} = \sqrt{\sum_{k=1}^{N} \mathbf{1}[y_i \neq y_k]} \; \frac{\exp\left(\frac{s_{ij}}{\tau}\right) \mathbf{1}_{[y_i \neq y_j]}}{\sum_{k=1}^{N} \exp\left(\frac{s_{ik}}{\tau}\right) \mathbf{1}_{[y_i \neq y_k]} + \varepsilon} \tag{7}$$

where τ represents the temperature hyperparameter.

D. Curriculum Learning

Humans and animals tend to learn better when training data are presented or organized in a meaningful sequence, rather than in random order, and gradually learn from more difficult to simple [24]. Such learning patterns are called curriculum learning. In our context, we propose a learning strategy that uses intra-class uncertainty to help the model learn meaningful semantic features progressively. Uncertainty from low similarity of same-class samples tends to show semantic discriminability, where sample pairs characterized by a lower intra-class uncertainty are those that show much higher semantic closeness. The progressive learning methodology used in our strategy is inspired by Chang et al. [25], where smaller weights are attached to positive pairs (the same class instances) with greater uncertainty in the early phase of training and progressively reverse this assignment throughout the training process. During curriculum learning, the model initially pays attention to easy instances pairs and then gradually shifts attention towards hard pairs. In this way, the knowledge provided by positive pairs is fully used, while the opposite effects are reduced. So, the progressive uncertainty aware function is defined as follows Given an uncertainty matrix $u \in \mathbb{R}^{N \times M}$, for each element u_{ij}, we define:

$$\theta_{ij}(u) = 1 + \tanh\left(\frac{\text{epoch}(i)}{T}) \cdot (\phi(\rho(u_{ij}))\right) \tag{8a}$$

$$\phi(\rho(u_{ij})) = \frac{\text{rank}(u_{ij}) - 1}{M - 1} \tag{8b}$$

$\text{rank}(u_{ij})$ denotes the descending rank of u_{ij} within its row, T (total number of epochs), $\text{epoch}(i)$ denotes the current epoch corresponding to iteration i

E. Loss Function

After coming across how a model can learn to be aware of uncertainty related to low intra-class similarity and mitigate high inter-class similarity, we define the learning loss function $\mathcal{L}$, P_i the set of positive samples, for instance i all indices j whose labels are the same as y_j, excluding i itself, N_i the set of negative samples, for instance i: all indices k whose labels are different from yi, excluding i itself, V is a set of all valid samples i for which P_i is non-empty and l_i is the contrastive loss for anchor i.

through a combination of Eq. (1b), (5), (6), (7), and (8a) as follows:

$$\mathcal{L} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \ell_i, \quad \mathcal{V} = \{i \mid |P_i| > 0\}, \tag{9}$$

$$\ell_i = -\frac{1}{|P_i|} \sum_{j \in P_i} \log \frac{w_{ij}\, e^{s_{ij}}}{w_{ij}\, e^{s_{ij}} + \sum\limits_{k \in N_i} \alpha e^{s_{ik}} + \varepsilon}, \tag{10}$$

$$P_i = \{j \mid y_j = y_i, j \neq i\}, \qquad N_i = \{k \mid y_k \neq y_i, k \neq i\}, \tag{11}$$

Algorithm 1 Learning Algorithm

Require: Training dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, encoder $f(\cdot)$, projection head $g(\cdot)$, epochs E, cohesiveness c_{ij}, non-cohesiveness c_{ij}, batch size M, temperature τ
Ensure: Trained encoder parameters Θ

```
Initialize encoder f(·) and projection head g(·)
Input preparation: dataset D = {(x_i, y_i)}_{i=1}^{N}
for epoch = 1 to E do
    for each mini-batch B = {(x_i, y_i)}_{i=1}^{M} do
        Data augmentation
        x̃_i^(1), x̃_i^(2) ← t(x_i)
        Feature extraction
        h_i^(k) ← f(x̃_i^(k))
        Projection
        z_i^(k) ← g(h_i^(k))
        Normalize embeddings:
        z_i^(k) ← z_i^(k) / ||z_i^(k)||
        Similarity computation
        s_ij ← z_i^⊤ z_j / τ
        Intra-class uncertainty
        c_ij ← exp(softmax_j(s_ij))
        u_ij ← W / (c_ij + c̄_ij + W)  Eq.(5)
        Progressive uncertainty weighting
        w_ij ← θ_ij(u_ij)  Eq.(6)
        Hard negative mining
        α_ij ← exp(s_ij/τ) 1[y_i ≠ y_j] / (Σ_{k=1}^{M} exp(s_ik/τ) 1[y_i ≠ y_k] + ε)  Eq.(7)
        Loss computation
        L ← (1/M) Σ_{i=1}^{M} ℓ_i(α_i, w_ij)  Eq.(9)
        Parameter update
        Θ ← Θ − η∇_Θ L
    end for
end for
return Θ
```

IV. Experiments

A. Experimental setup

1) Dataset: To evaluate the effectiveness of the proposed ViCo-LWIR framework, experiments were conducted on a thermal image dataset containing 15,000 samples acquired using a long-wave infrared (LWIR) thermal sensor under controlled indoor conditions. Data collection was carried out at ambient temperatures ranging from (25°C) to (35°C), allowing the capture of thermal variations encountered in typical indoor environments. Each sample corresponds to a single thermal frame extracted from static scene recordings

or short capture sequences. The dataset includes multiple spill acquisition scenarios designed to simulate realistic liquid dispersion patterns. Spill samples were generated by pouring liquids in controlled quantities onto different surface types, including dark matte floors and reflective surfaces, while maintaining fixed camera viewpoints. To capture a wide range of spill appearances, variations in the poured volume, release location, liquid temperature, and spreading dynamics were introduced. Since thermal appearance depends on temperature contrast with the surrounding environment, warm spills typically appear as bright regions, whereas colder spills appear darker. These variations resulted in diverse spill morphologies, including large puddles, thin streaks, irregular boundaries, and fragmented droplets. Additional environmental diversity was incorporated through variations in background texture, object placement, partial occlusions, and minor viewpoint changes. Human interaction was also included in some scenarios, where hands appeared near or during spill events, increasing realism and introducing challenging occlusion conditions. The no-spill class was collected under identical acquisition settings without liquid introduction. These samples include clean surfaces as well as hard negative cases containing thermally or visually confusing patterns such as reflections, shadows, and textured regions, improving the robustness of the learned representations. The dataset was divided into training and validation sets using a stratified 90% and 10% split performed at the scene level to prevent information leakage between splits. Due to privacy and data-sharing constraints, the full dataset cannot be publicly released. Instead, we provide a separate benchmark test set containing 2,000 samples acquired under conditions intentionally excluded from the training and validation distributions. This out of distribution test set includes spills on previously unseen surfaces such as tables, fabric garments (shirts), and wooden materials. By evaluating on surfaces and environments not observed during training, the benchmark enables a rigorous assessment of the framework's ability to generalize beyond the acquisition conditions used for model development.

2) Implementation details.: A pretrained Vision Transformer base model (*ViT-B/16*) [26] was employed as the backbone network. A multilayer perceptron (MLP) projector head with an output dimension of 256 was appended to adapt the model to the contrastive learning framework. Fine-tuning was performed using patch-level representations from (*ViT-B/16*) to capture fine-grained spatial details, as spills typically occur on a small scale, AdamW [27] optimizer was used with learning rates 9×10^{-5} for the projector head and 1×10^{-4} for the backbone. Training was carried out using a batch size of 64 on a A100 GPU equipped with 40 GB of memory. A temperature parameter of 0.1 was adopted, slightly higher than the base temperature (0.07) used by Khosla et al. [11]. Prior to training, all thermal images were normalized to the range $[0, 1]$ using a dataset mean of 0.24 and a standard deviation of 0.07. This normalization improves optimization stability and ensures consistent feature scaling across samples. Following pre-training, the backbone was frozen, and a classification head was introduced for the downstream classification task.

3) Evaluation metrics: To understand the underlying behavior of the embeddings within their latent representation space, we analyze the learned representation spaces using embedding analysis metrics introduced by Mildenberger et al. [28] where the concept of sample alignment accuracy (SAA) is introduced to determine whether the embeddings of positive pairs are more closely aligned with each other compared to other samples. Class alignment consistency (CAC) is the second metric introduced to measure how pure embedding neighborhoods are with respect to the latent class. Additionally, we used metrics inherit from traditional classification metrics, which are the F1 score (F1), Accuracy (ACC), Average precision (AP), and Recall (REC) to evaluate ViCo-LWIR performance.

B. Ablation study

In this section, we conduct an ablation study to evaluate the contribution of each component of the proposed framework. Four degraded variants are constructed by removing or modifying specific elements of the training objective. *Hard-Negative* Weighting denotes the model whose loss ℓ_i retains only the hard-negative weighting mechanism while removing uncertainty-based weighting. Only Uncertainty refers to the model that directly uses the estimated uncertainty without the progressive weighting function. *Uncertainty Curriculum Learning* denotes the model that employs only the progressive uncertainty weighting strategy for positive samples. *SupCon* [11] corresponds to the conventional supervised contrastive learning baseline. Finally, *All Components* represents the complete model trained by integrating uncertainty weighting, progressive curriculum learning, and hard-negative weighting.

The embedding visualizations further reveal clear differences in convergence behavior across the ablation variants in Fig. 3. The *Hard-Negative Weighting* model primarily emphasizes confusing negative samples by assigning larger weights to hard negatives. While this encourages stronger inter-class repulsion, it does not provide a mechanism for stabilizing positive-pair learning. Consequently, the optimization process tends to focus excessively on separating difficult negatives, leading to unstable feature organization and incomplete convergence. The resulting embeddings are more visually separated than those of other ablation models; however, their overall performance remains inferior to the complete framework. As shown in Table I, emphasizing hard negative samples alone is insufficient to achieve optimal performance. Similarly, both *Only Uncertainty* and *Uncertainty Curriculum Learning* improve the treatment of positive samples by accounting for uncertainty, but neither is sufficient to produce well-structured feature clusters on its own. Although these variants reduce the influence of unreliable samples and partially improve intra-class compactness, they lack the complementary negative-sample guidance required to achieve strong class separation. As a result, noticeable overlap between classes remains, and the learned representations fail to converge to highly discriminative clusters.

In contrast, the full model (*All Components*) is the only variant that achieves clear convergence and well-separated class

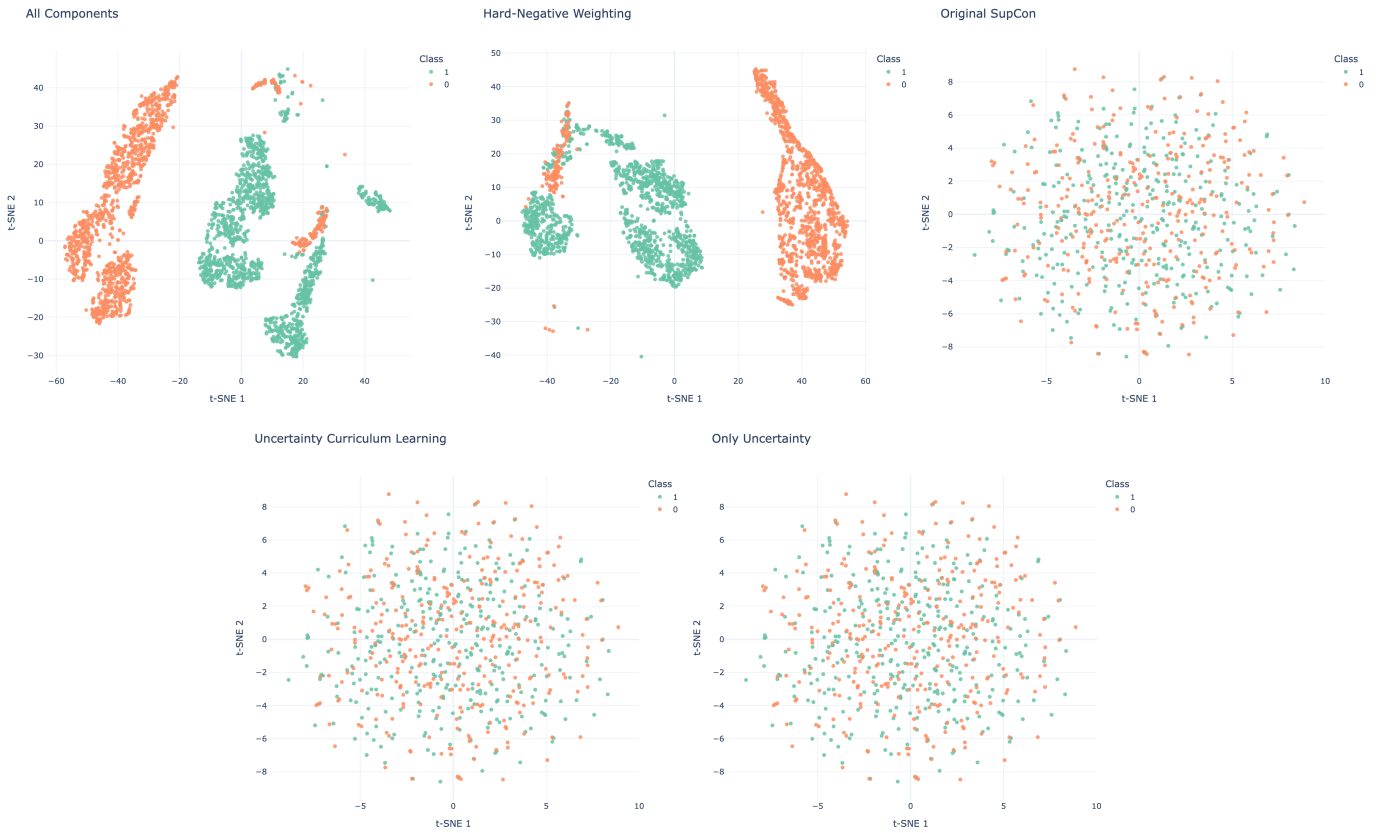


Fig. 3. The figure shows that the full framework produces well-separated t-SNE clusters, with compact intra-class structure and strong separation between non-spill (class 0) and spill (class 1) samples. In contrast, all ablation variants (*SupCon*, *Uncertainty Curriculum Learning*, and *Uncertainty-only*) show significant overlap, indicating weaker feature discrimination. *Hard-Negative Weighting* slightly improves separation but remains inferior to the *All Components*. Overall, the results show that combining uncertainty-aware positive selection with hard-negative mining is necessary for robust and well-separated representations.

distributions. The uncertainty-aware curriculum progressively introduces more challenging positive samples, improving intra-class compactness in a stable manner, while hard-negative weighting simultaneously strengthens inter-class discrimination by emphasizing confusing negatives. The combination of these mechanisms provides a balanced optimization process that promotes both class compactness and class separability. This synergy leads to significantly more discriminative feature representations and explains the superior performance of the complete framework.

The training loss curves shown in Fig 4 provide further insight into the contribution of each component of the proposed loss function, ℓ_i. Clear differences in optimization behavior can be observed across the ablation variants. The *Hard-Negative Weighting* variant initially achieves a rapid reduction in loss during the early training stages; however, the optimization quickly reaches a plateau during the middle-to-late epochs, indicating limited convergence. This behavior suggests that emphasizing difficult negative samples alone is insufficient to sustain continued representation refinement throughout training. The *Uncertainty Curriculum Learning* variant exhibits a different trend. Although the loss decreases during the initial epochs, it subsequently oscillates around a relatively high value and remains trapped in a suboptimal region of the loss landscape. This instability indicates that gradually introducing uncertain positive samples without complementary negative-sample guidance does not provide enough discriminative information to achieve effective convergence Similarly, the *Only Uncertainty* variant demonstrates a gradual decrease in loss during the early training phase, reaching a moderate improvement over the initial value. However, the optimization process eventually stagnates and partially regresses toward higher loss values, suggesting that uncertainty-aware positive-pair weighting alone cannot maintain stable long-term optimization. The *Original SupCon* baseline exhibits the slowest convergence among all variants. Although the loss decreases consistently throughout training, the reduction is limited and progresses at a significantly slower rate compared to the proposed method. This observation highlights the inherent limitations of standard supervised contrastive learning when dealing with ambiguous or difficult thermal image samples.

In contrast, the complete framework (*All Components*) achieves the fastest and most stable convergence. The training loss decreases consistently throughout the optimization process and reaches the lowest final value among all compared methods. This behavior demonstrates that the combination of uncertainty-aware curriculum learning and hard-negative weighting provides complementary benefits, resulting in a more effective and stable optimization process.

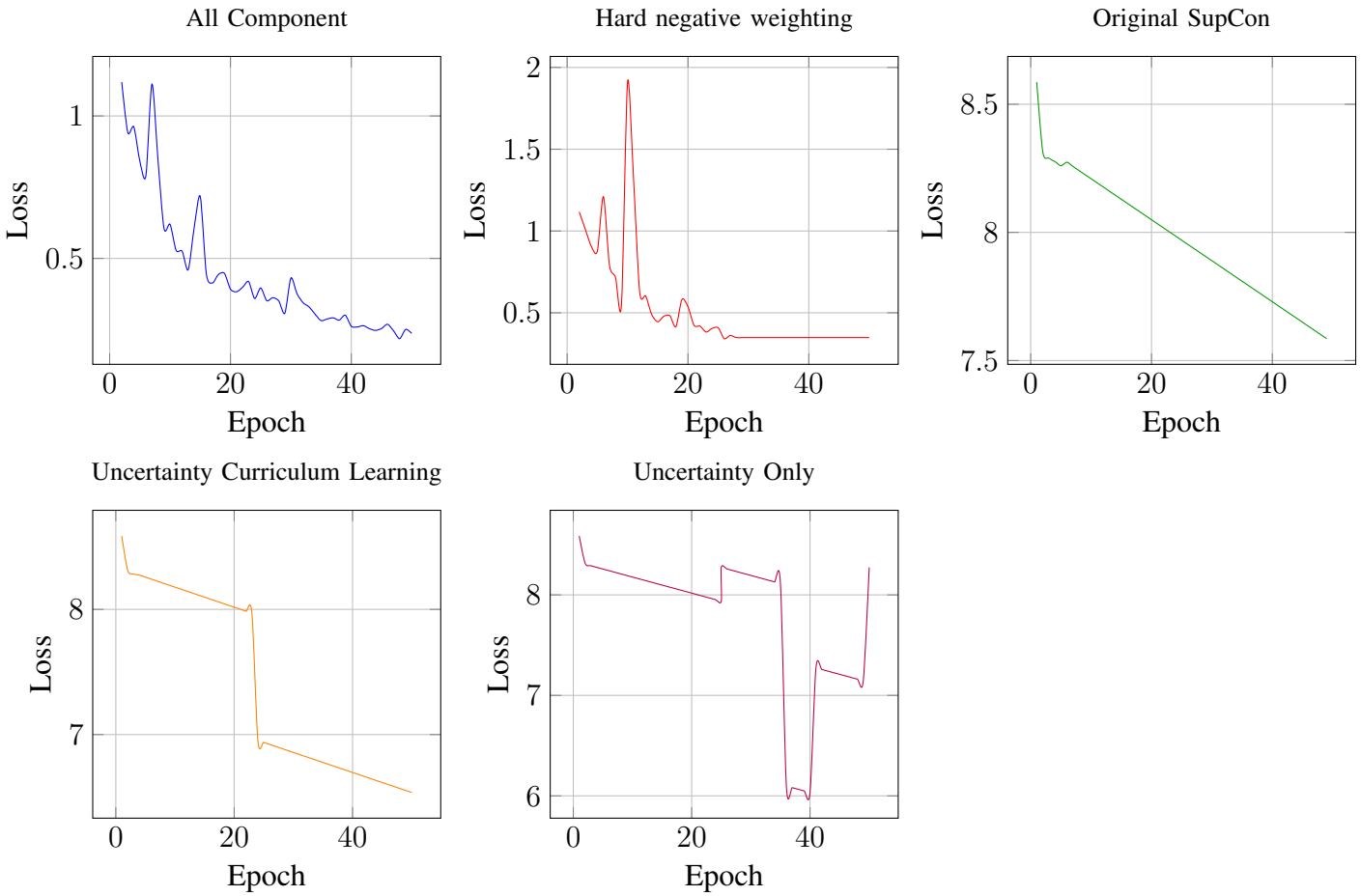


Fig. 4. This figure compares training loss across ablated model variants and the full ViCo-LWIR framework. Each variant removes a key component, while the full model includes all components. The results show that the complete framework converges faster and more stably, demonstrating the effectiveness of each component in improving optimization performance.

TABLE I
ABLATION STUDY OF THE PROPOSED VICO-LWIR FRAMEWORK. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER 3 INDEPENDENT RUNS, AND ALL VALUES CORRESPOND TO TOP-1 ACCURACY (%).

Ablation Component	**ACC (%)**	**AP (%)**	**REC (%)**	**F1 (%)**
Baseline (SupCon)	59.00 ± 0.24	51.10 ± 1.22	59.00 ± 0.24	51.90 ± 0.33
+ Hard-Negative Weighting	$59.016, \pm 0.42$	51.49 ± 1.66	59.0 ± 0.42	51.34 ± 0.57
+ Uncertainty Curriculum Learning	63.0 ± 0.24	51.24 ± 1.25	59.00 ± 0.24	51.0 ± 0.33
+ Only Uncertainty	58.80 ± 0.24	51.68 ± 0.55	58.00 ± 0.24	51.79 ± 0.18
+ All Components (full Framework)	97.24 ± 0.06	96.05 ± 0.06	95.8 ± 0.08	96.08 ± 0.08

C. Comparison with State-of-the-Art Contrastive Learning Methods

This section compares the proposed ViCo-LWIR framework with existing weakly supervised contrastive learning methods, highlighting improvements in representation quality and classification performance. We also analyze Sample Alignment Accuracy (SAA) and Class Alignment Consistency (CAC) over 40 training epochs.

SAA Fig. 5 measures the proportion of samples whose nearest neighbors in the embedding space belong to the same class, providing an indicator of local representation consistency. Higher SAA values indicate stronger semantic alignment and improved intra-class compactness. The consistently high SAA observed throughout training demonstrates that the learned representations remain well-aligned and discriminative, enabling reliable separation of spill and non-spill classes.

CAC Fig 6 quantifies the stability of class-wise feature representations during training by measuring how consistently samples from the same class remain aligned in the embedding space. Higher CAC values reflect stronger intra-class compactness and more stable semantic structure. The results show consistently high CAC across training epochs, with only minor fluctuations, indicating that the model maintains coherent and robust class-wise grouping while optimizing the contrastive objective, leading to stable and discriminative feature learning.

TABLE II
PERFORMANCE METRICS OF VARIOUS CONTRASTIVE LEARNING FRAMEWORKS ON VICO-LWIR USING A VISION TRANSFORMER BACKBONE (VIT-B). RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER 3 INDEPENDENT RUNS, AND ALL VALUES CORRESPOND TO TOP-1 ACCURACY (%).

Method	Arch.	ACC	AP	REC	F1
ViCo-LWIR (ours)	ViT-B	97.24 ± 0.06	96.05 ± 0.06	95.8 ± 0.08	96.08 ± 0.08
SCHaNe [29]	ViT-B	63.28 ± 0.80	67.34 ± 0.75	51.56 ± 0.90	58.40 ± 0.85
VarSupCon [30]	ViT-B	78.59 ± 0.30	84.12 ± 0.28	82.81 ± 0.32	83.46 ± 0.30
SCPL [31]	ViT-B	86.78 ± 0.25	80.75 ± 0.20	84.37 ± 0.30	87.44 ± 0.27
WSCL-TL [32]	ViT-B	69.81 ± 0.70	70.37 ± 0.65	95.62 ± 0.20	78.13 ± 0.50

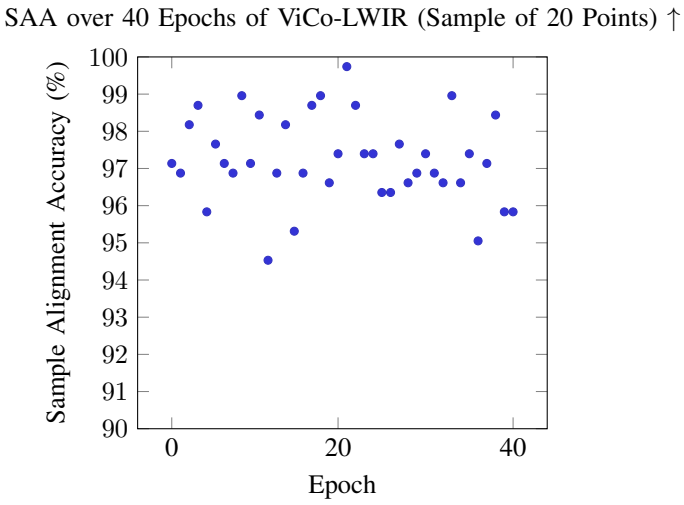


Fig. 5. Sample Alignment Accuracy (SAA) across 50 training epochs of ViCo-LWIR. SAA measures the proportion of samples whose nearest neighbors in the embedding space belong to the same class, providing an indicator of local representation consistency. Higher SAA values correspond to stronger semantic alignment and improved intra-class compactness. The consistently high SAA observed throughout training demonstrates that the learned representations remain well aligned and discriminative, enabling reliable separation of spill and non-spill classes.

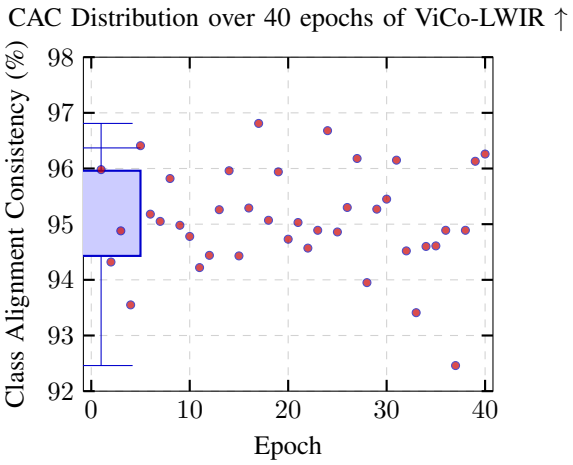


Fig. 6. Class Alignment Consistency (CAC) over 40 epochs measures how stably samples from the same class remain grouped in the embedding space during training. High CAC indicates strong intra-class compactness and stable semantic structure. The consistently high and stable CAC values, with only minor fluctuations, show that the model maintains coherent class-wise grouping throughout training, producing robust and discriminative representations.

D. Vico-LWIR industrial setting

The ViCo-LWIR framework is designed for real-world industrial thermal spill detection, leveraging thermal imaging since many liquid spills are transparent and therefore difficult to detect using RGB modalities. A dedicated thermal camera setup is used for data collection, enabling robust representation learning through fine-tuning on thermal inputs captured under realistic operating conditions. The learned representations further generalize across different surface types and material properties, as illustrated in Fig 7, demonstrating strong adaptability to variations in both thermal response and environmental context. This cross-surface generalization is particularly important in industrial scenarios where spills may occur on diverse substrates with differing thermal characteristics. The evaluation is conducted on a dataset out of the distribution of training and validation dataset, ensuring a reliable assessment of generalization performance and robustness under realistic data shifts. Finally, the trained model can be deployed on edge devices with minimal GPU requirements, enabling efficient and low-latency real-time inference in practical industrial environments.

V. CONCLUSION

In this work, we present a vision-based system that employs a contrastive learning framework using LWIR imaging for spill detection. The proposed approach effectively reduces annotation costs through weak supervision, while producing generalizable representations capable of detecting spills despite irregular shapes and class-level variability, as demonstrated by the model's performance across the training data distribution. Moreover, the system enables precise measurement and monitoring of spill characteristics, demonstrating its applicability in instrumentation settings where accurate quantification and reliable detection are critical for operational decision-making.

However, ViCo-LWIR exhibits limitations in multi-instance scenarios. This behavior reflects a common challenge in weakly supervised learning, where limited annotation signals often lead to incomplete instance coverage, particularly in complex or crowded environments. This leads us to consider future research directions in the forthcoming paper, Instruction-Tuned Contrastive Learning for generalizable spill detection under class-level variability. A multimodal that leverages RGB-D-T alignment and instruction-based supervision with text, the model learns where to focus in complex multi-instance scenes, enhancing generalization across diverse spill types and environments.

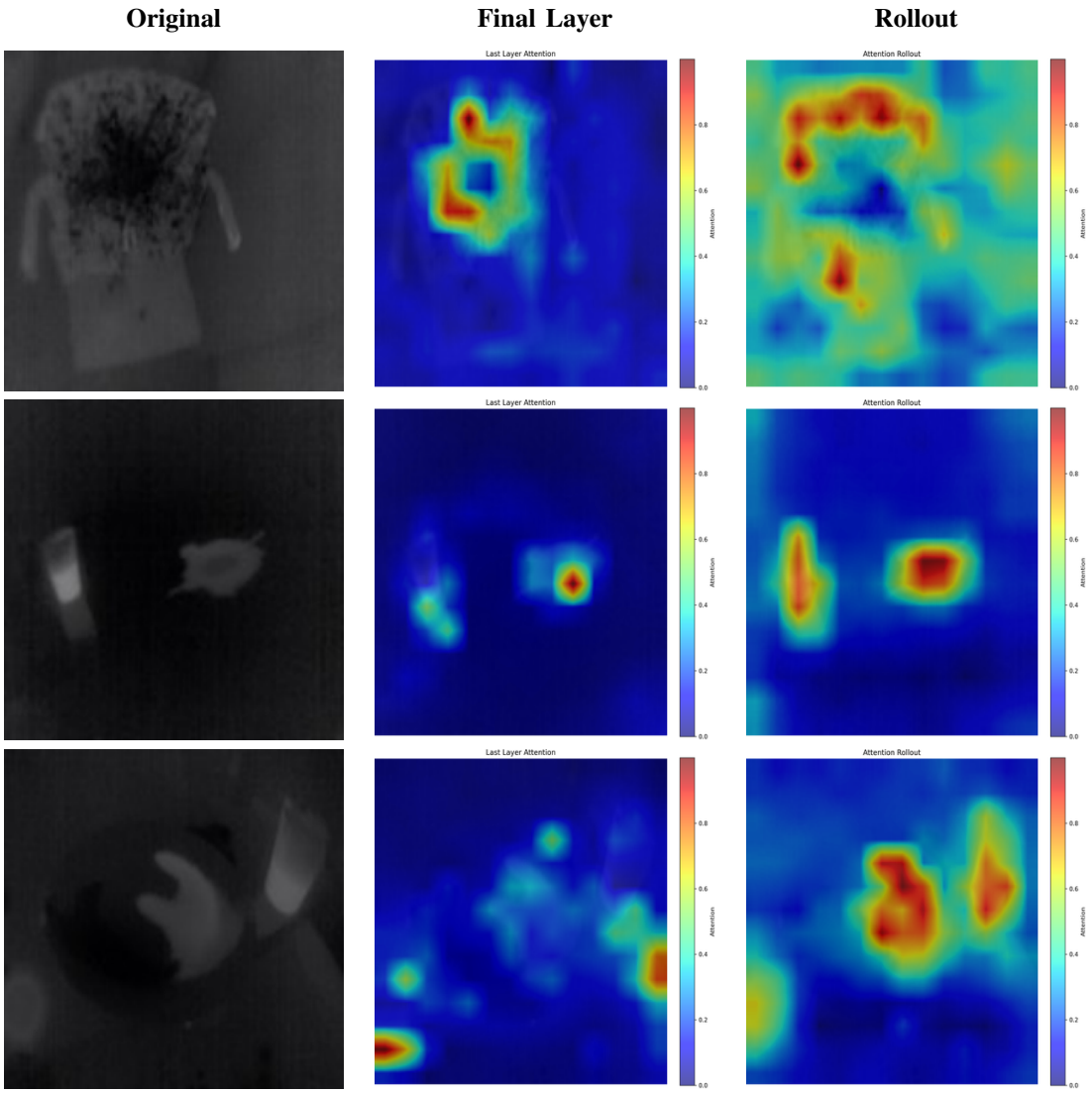


Fig. 7. Comparison of original images, final-layer attention maps, and attention rollout visualizations.

REFERENCES

[1] N. Lalam, H. Bhatta, S. Bukka, P. Zhang, N. Diemler, A. Shumski, P. Ohodnicki, M. Buric, and R. Wright, "Deep neural network assisted distributed strain and temperature fiber sensor system for natural gas pipeline monitoring," *IEEE Transactions on Instrumentation and Measurement*, 2025.

[2] J. Ma, L. Tang, M. Xu, H. Zhang, and G. Xiao, "Stdfusionnet: An infrared and visible image fusion network based on salient target detection," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–13, 2021.

[3] S. Vidas, R. Lakemond, S. Denman, C. Fookes, S. Sridharan, and T. Wark, "A mask-based approach for the geometric calibration of thermal-infrared cameras," *IEEE Transactions on Instrumentation and Measurement*, vol. 61, no. 6, pp. 1625–1635, 2012.

[4] V. Bherde, U. Balunaini, B. V. N. P. Kambhammettu, and S. Regonda, "Infrared thermography based seepage and erosion detection in earthen dams using laboratory scale models," *Scientific Reports*, vol. 15, 2025.

[5] H. Wang, Y. Wang, Y. Zhu, M. Zhang, T. Yan, and X. Niu, "Infrared thermography for leakage detection in underground water conveyance tunnels," *Measurement*, vol. 258, Jan. 2026.

[6] G. Rouhafzay, S. Rowlands, A. J. Valencia, S. Yang, P. Payeur, H. Tian, and J. Dickens, "Multi-spectral imaging and data fusion for real-time bleeding detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. IEEE, 2025, pp. 4600–4607.

[7] T.-W. Ke, J.-J. Hwang, and S. X. Yu, "Universal weakly supervised segmentation by pixel-to-segment contrastive learning," *arXiv preprint arXiv:2105.00957*, 2021.

[8] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *International conference on machine learning*. PmLR, 2020, pp. 1597–1607.

[9] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 9729–9738.

[10] C.-Y. Chuang, J. Robinson, Y.-C. Lin, A. Torralba, and S. Jegelka, "Debiased contrastive learning," *Advances in neural information processing systems*, vol. 33, pp. 8765–8775, 2020.

[11] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised contrastive learning," *Advances in neural information processing systems*, vol. 33, pp. 18 661–18 673, 2020.

[12] J. Li, P. Zhou, C. Xiong, and S. C. Hoi, "Prototypical contrastive learning of unsupervised representations," *arXiv preprint arXiv:2005.04966*, 2020.

[13] G. Rouhafzay, S. Rowlands, A. J. Valencia, S. Yang, P. Payeur, H. Tian, and J. Dickens, "Multi-spectral imaging and data fusion for real-time bleeding detection," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4561–4568.

[14] G. Rouhafzay, H. Tian, and P. Payeur, "Warm liquid spill detection and tracking using thermal imaging," in *2022 IEEE 9th International*

Conference on Computational Intelligence and Virtual Environments for Measurement Systems and Applications (CIVEMSA). IEEE, 2022, pp. 1–5.

[15] J. Seo, W. Bae, D. J. Sutherland, J. Noh, and D. Kim, "Object discovery via contrastive learning for weakly supervised object detection," in *European conference on computer vision*. Springer, 2022, pp. 312–329.

[16] Y. Zhang, C. Zhu, G. Yang, and S. Chen, "Negative prototypes guided contrastive learning for weakly supervised object detection," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2023, pp. 36–51.

[17] A. Brea, F. J. García-Corbeira, E. Tsiranidou, G. C. Peláez, L. Díaz-Vilariño, and J. Martínez, "Low-cost thermal point clouds of indoor environments," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 48, pp. 99–105, 2024.

[18] FLIR Systems, "Lepton," https://oem.flir.com/products/lepton, 2026, [Online]; Accessed: May 22, 2026.

[19] M. Reichstein, G. Camps-Valls, B. Stevens, M. Jung, J. Denzler, N. Carvalhais, and F. Prabhat, "Deep learning and process understanding for data-driven earth system science," *Nature*, vol. 566, no. 7743, pp. 195–204, 2019.

[20] W. He, Z. Jiang, T. Xiao, Z. Xu, and Y. Li, "A survey on uncertainty quantification methods for deep learning," *arXiv preprint arXiv:2302.13425*, 2023.

[21] M. Valdenegro-Toro, "I find your lack of uncertainty in computer vision disturbing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1263–1272.

[22] A. Jsang, *Subjective Logic: A formalism for reasoning under uncertainty*. Springer Publishing Company, Incorporated, 2018.

[23] A. Jøsang, "Belief mosaics of subjective opinions," in *2019 22th International Conference on Information Fusion (FUSION)*. IEEE, 2019, pp. 1–8.

[24] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum learning," in *Proceedings of the 26th annual international conference on machine learning*, 2009, pp. 41–48.

[25] L. Chang, L. Chen, and C. Zhou, "Uncertainty-aware contrastive learning for deep clustering," *Neurocomputing*, p. 130568, 2025.

[26] A. Dosovitskiy, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[27] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017.

[28] D. Mildenberger, P. Hager, D. Rueckert, and M. J. Menten, "A tale of two classes: Adapting supervised contrastive learning to binary imbalanced datasets," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 10 305–10 314.

[29] Z. Long, G. Killick, R. McCreadie, G. A. Camarasa, and Z. Meng, "When hard negative sampling meets supervised contrastive learning," *arXiv preprint arXiv:2308.14893*, 2023.

[30] Z. Wang, J. Fan, T. Nguyen, H. Ji, and G. Liu, "Variational supervised contrastive learning," *arXiv preprint arXiv:2506.07413*, 2025.

[31] I. Fostiropoulos and L. Itti, "Supervised contrastive prototype learning: Augmentation free robust neural network," *arXiv preprint arXiv:2211.14424*, 2022.

[32] M. He, L. Qin, Y. Wang, X. Deng, Q. Liu, Y. Zhang, and K. Liu, "A weakly supervised contrastive learning pre-training method for visual defect detection of transmission lines," *IEEE Transactions on Instrumentation and Measurement*, 2025.